

	(c)	line: reasonable shape with less ripple.....	B1	[1]
5	(a)	field producing force of 1.0 N m^{-1} on wire <i>OR</i> $B = F/IL\sin\theta$ carrying current of 1.0 A normal to field <i>OR</i> symbols explained ...	M1 A1	[2]
	(b)	(i) $\phi = BA$ $= 1.8 \times 10^{-4} \times 0.60 \times 0.85$ $= 9.18 \times 10^{-5} \text{ Wb}$	C1 A1	[2]
		(ii)1 $\Delta\phi = 9.18 \times 10^{-5} \text{ Wb}$	A1	
		(ii)2 $e = (N\Delta\phi)/\Delta t$ $= (9.18 \times 10^{-5})/0.20$ $= 4.59 \times 10^{-4} \text{ V}$	C1 A1	[3]
		(iii) there is an e.m.f. and a complete circuit <i>OR</i> no resultant e.m.f. from other three sides <i>OR</i> no e.m.f. in AB so yes	B1	[1]
6	(a)	packet/quantum of energy.....	M1	